

Q1 (1)

$$\left(3^{1/4} + 5^{1/8}\right)^{60}$$

$${}^{60}C_r \left(3^{1/4}\right)^{60-r} \cdot \left(5^{1/8}\right)^r$$

$${}^{60}C_r (3)^{\frac{60-r}{4}} \cdot 5^{\frac{r}{8}}$$

For rational terms.

$$\frac{r}{8} = k; 0 \leq r \leq 60$$

$$0 \leq 8k \leq 60$$

$$0 \leq k \leq \frac{60}{8}$$

$$0 \leq k \leq 7.5$$

 $k = 0, 1, 2, 3, 4, 5, 6, 7$ $\frac{60-8k}{4}$ is always divisible by 4 for all value of k Total rational terms = 8 Total terms = 61

irrational terms = 53

$$n - 1 = 53 - 1 = 52$$

52 is divisible by 26.

Q2 (3)

$$(1 - x + x^3)^n = \sum_{j=0}^{3n} a_j x^j$$

$$(1 - x + x^3)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{3n} x^{3n}$$

$$\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} = \text{Sum of } a_0 + a_2 + a_4 + \dots$$

$$\sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1} = \text{Sum of } a_1 + a_3 + a_5 + \dots$$

put $x = 1$

$$1 = a_0 + a_1 + a_2 + a_3 + \dots + a_{3n} \quad \dots (A)$$

Put $x = -1$

Hints and Solutions

MathonGo

$$1 = a_0 - a_1 + a_2 - a_3 + \dots + (-1)^{3n} a_{3n} + \dots$$

Solving (A) and (B)

$$a_0 + a_2 + a_4 + \dots = 1$$

$$a_1 + a_3 + a_5 + \dots = 0$$

$$\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} + 4 \sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1} = 1$$

Q3 (6)

$$A = \sum_{k=0}^n {}^nC_k \left[\left(-\frac{1}{2}\right)^2 + \left(\frac{-3}{4}\right)^2 + \left(\frac{-7}{8}\right)^2 + \left(\frac{-15}{16}\right)^2 + \left(\frac{-31}{32}\right)^2 \right]$$

$$A = \left(1 - \frac{1}{2}\right)^2 + \left(1 - \frac{3}{4}\right)^2 + \left(1 - \frac{7}{8}\right)^2 + \left(1 - \frac{15}{16}\right)^2 + \left(1 - \frac{31}{32}\right)^2$$

$$A = \frac{1}{2^n} + \frac{1}{4^n} + \frac{1}{8^n} + \frac{1}{16^n} + \frac{1}{32^n}$$

$$A = \frac{1}{2^n} \left(\frac{1 - \left(\frac{1}{2^n}\right)^5}{1 - \frac{1}{2^n}} \right) \Rightarrow A = \frac{\left(1 - \frac{1}{2^{5n}}\right)}{(2^n - 1)}$$

$$(2^n - 1)A = 1 - \frac{1}{2^{5n}}, \text{ Given } 63A = 1 - \frac{1}{2^{30}}$$

Clearly $5n = 30$

$$n = 6$$

Q4 (1)

$${}^7C_3 x^4 x^{(3 \log_2^2)} = 4480$$

$$\Rightarrow x^{(4+3 \log_2^2)} = 2^7$$

$$\Rightarrow (4 + 3t)t = 7; t = \log_2^x$$

$$\Rightarrow t = 1, \frac{-7}{3} \Rightarrow x = 2$$

Q5 (4)

$$(2023 - 2)^{3762} = 2023k_1 + 2^{3762}$$

$$= 17k_2 + 2^{3762} \text{ (as } 2023 = 17 \times 17 \times 9)$$

$$= 17k_2 + 4 \times 16^{940}$$

$$= 17k_2 + 4 \times (17 - 1)^{940}$$

Hints and Solutions

MathonGo

$$= 17k_2 + 4(17k_3 + 1)$$

$$= 17k + 4 \Rightarrow \text{remainder} = 4$$

Q6 (4)

$$\begin{aligned} & \sum_{r=0}^6 {}^6C_r \cdot {}^6C_{6-r} \\ &= {}^6C_0 \cdot {}^6C_6 + {}^6C_1 \cdot {}^6C_5 + \dots + {}^6C_6 \cdot {}^6C_0 \end{aligned}$$

Now,

$$\begin{aligned} & (1+x)^6(1+x)^6 \\ &= ({}^6C_0 + {}^6C_1x + {}^6C_2x^2 + \dots + {}^6C_6x^6) \\ & \quad ({}^6C_0 + {}^6C_1x + {}^6C_2x^2 + \dots + {}^6C_6x^6) \end{aligned}$$

Comparing coefficient of x^6 both sides

$${}^6C_0 \cdot {}^6C_6 + {}^6C_1 \cdot {}^6C_5 + \dots + {}^6C_6 \cdot {}^6C_0 = {}^{12}C_6$$

$$= 924$$

Q7 (4)

$$\begin{aligned} T_{r+1} &= {}^nC_r (x)^{n-r} \left(\frac{a}{x^2} \right)^r \\ &= {}^nC_r a^r x^{n-3r} \end{aligned}$$

$${}^nC_2 a^2 : {}^nC_3 a^3 : {}^nC_4 a^4 = 12 : 8 : 3$$

After solving

$$n = 6, a = \frac{1}{2}$$

For term independent of 'x' $\Rightarrow n = 3r$

$$r = 2$$

$$\therefore \text{Coefficient is } {}^6C_2 \left(\frac{1}{2}\right)^2 = \frac{15}{4}$$

Nearest integer is 4

Q8 (2)

$$(1 + x + 2x^2)^{20} = a_0 + a_1x + \dots + a_{40}x^{40} \text{ put } x=1, -1$$

$$\Rightarrow a_0 + a_1 + a_2 + \dots + a_{40} = 2^{20}$$

$$a_0 - a_1 + a_2 - \dots + a_{40} = 2^{20}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{39} = \frac{4^{20} - 2^{20}}{2}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{37} = 2^{39} - 2^{19} - a_{39}$$

$$\text{here } a_{39} = \frac{20!(2)^{19} \times 1}{19!} = 20 \times 2^{19}$$

$$\Rightarrow a_1 + a_3 + \dots + a_{37} = 2^{19} (2^{20} - 1 - 20)$$

$$= 2^{19} (2^{20} - 21)$$

Q9 (210)

$$\left((x^{1/3} + 1) - \left(\frac{\sqrt{x+1}}{\sqrt{x}} \right) \right)^{10}$$

$$(x^{1/3} - x^{-1/2})^{10}$$

$$T_{r+1} = {}^{10}C_r (x^{1/3})^{10-r} (-x^{-1/2})^r$$

$$\frac{10-r}{3} - \frac{r}{2} = 0 \Rightarrow 20 - 2r - 3r = 0$$

$$\Rightarrow r = 4$$

$$T_5 = {}^{10}C_4 = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$$

Q10 (19)

Instead of nC_k it must be ${}^{10}C_k$ i.e.

$$\sum_{k=0}^{10} (2^2 + 3k) {}^{10}C_k = \alpha \cdot 3^{10} + \beta \cdot 2^{10}$$

Hints and Solutions

MathonGo

$$\text{LHS} = 4 \sum_{k=0}^{10} {}^{10}C_k + 3 \sum_{k=0}^{10} k \cdot \frac{10}{k} \cdot {}^{9}C_{k-1}$$

$$= 4 \cdot 2^{10} + 3 \cdot 10 \cdot 2^9$$

$$= 19 \cdot 2^{10} = \alpha \cdot 3^{10} + \beta \cdot 2^{10}$$

$$\Rightarrow \alpha = 0, \beta = 19 \Rightarrow \alpha + \beta = 19$$

Q11 (38)

$$\text{Equation of plane is } \begin{vmatrix} x-1 & y+6 & z+5 \\ 3 & 4 & 2 \\ 4 & -3 & 7 \end{vmatrix} = 0$$

, α) lies on it so Now $(1, -1,$

$$\begin{vmatrix} 0 & 5 & \alpha+5 \\ 3 & 4 & 2 \\ 4 & -3 & 7 \end{vmatrix} = 0 \Rightarrow 5\alpha + 38 = 0 \Rightarrow 15\alpha = 38$$